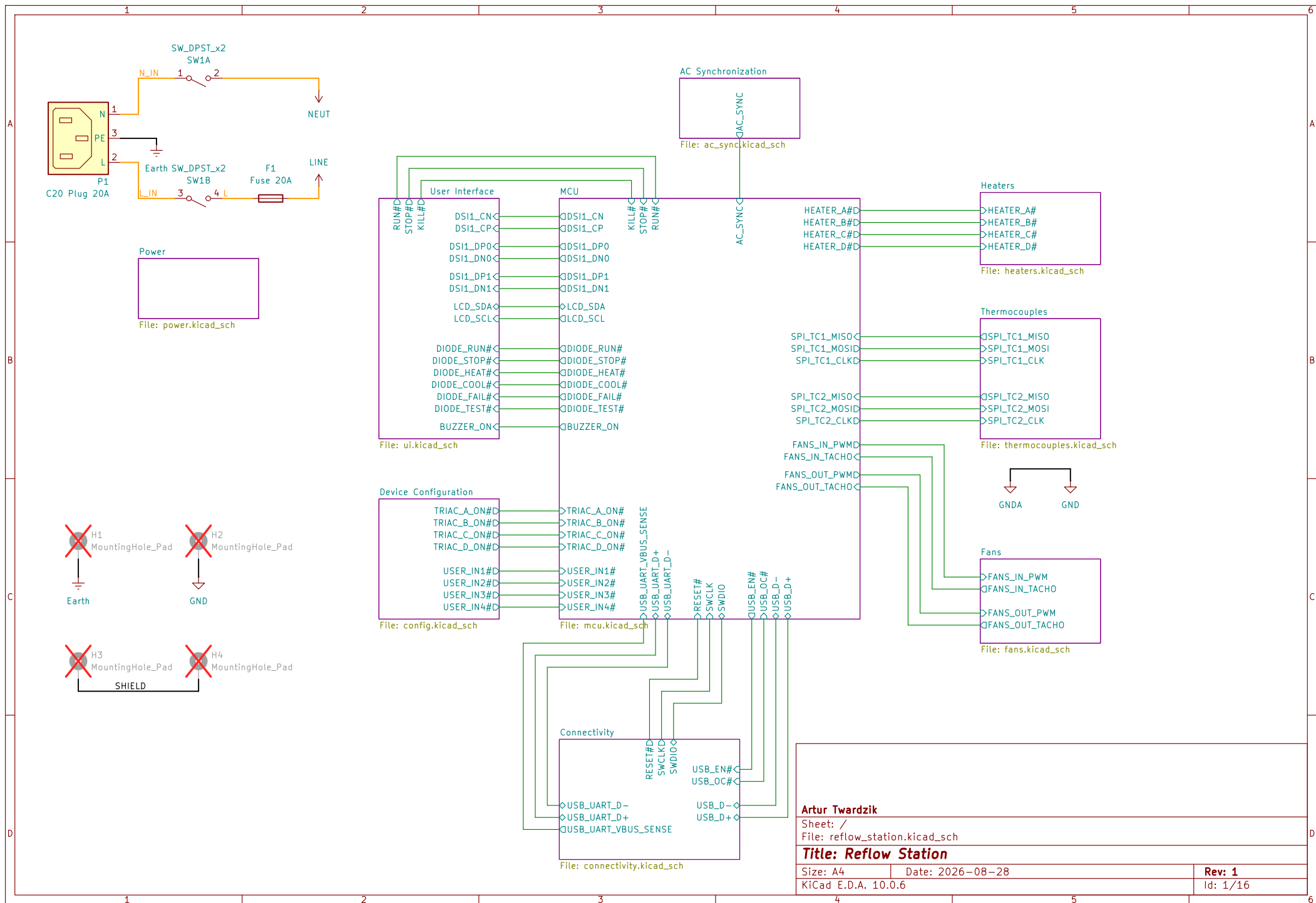
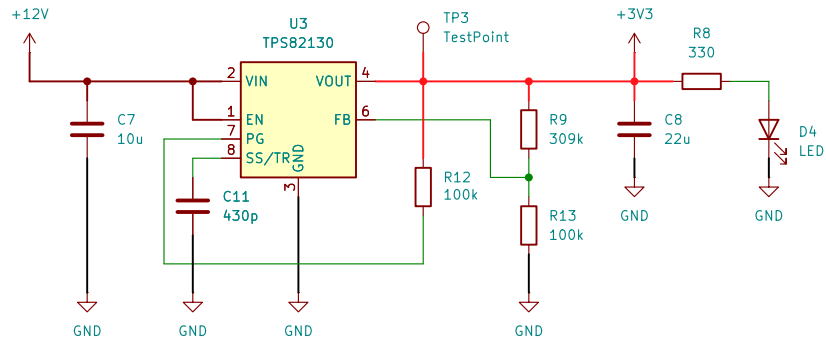
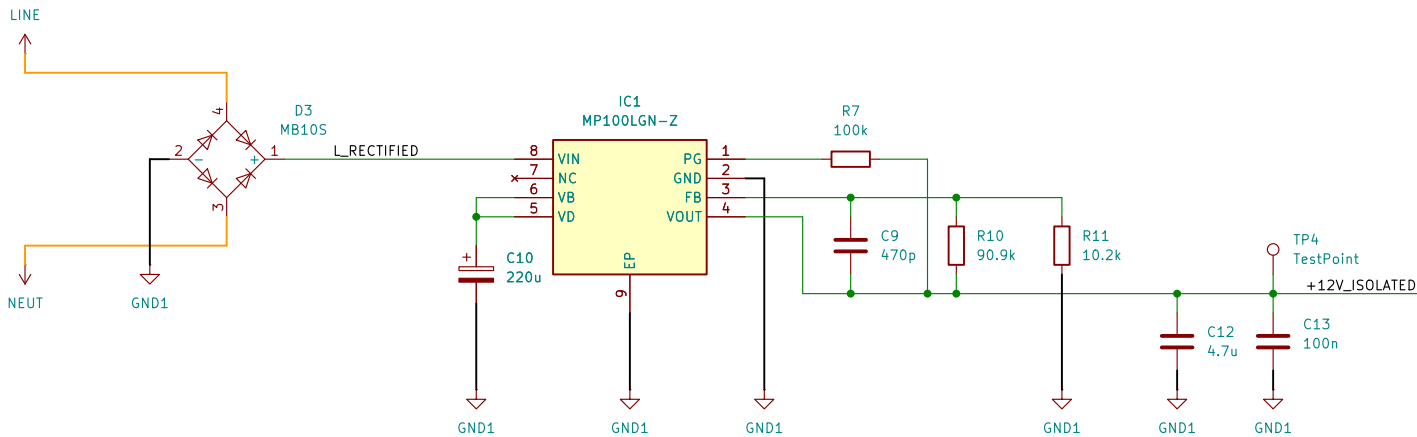
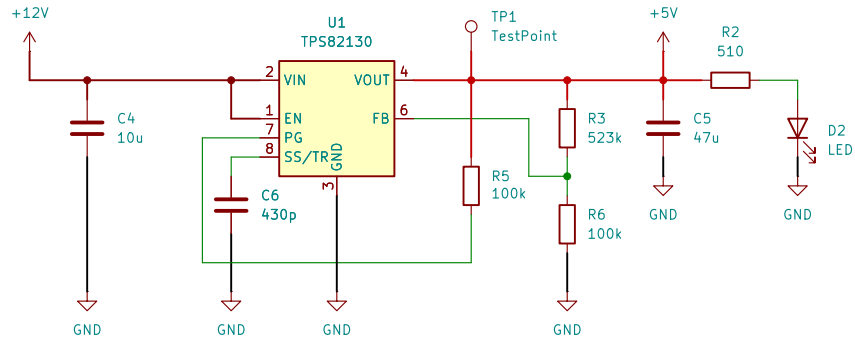
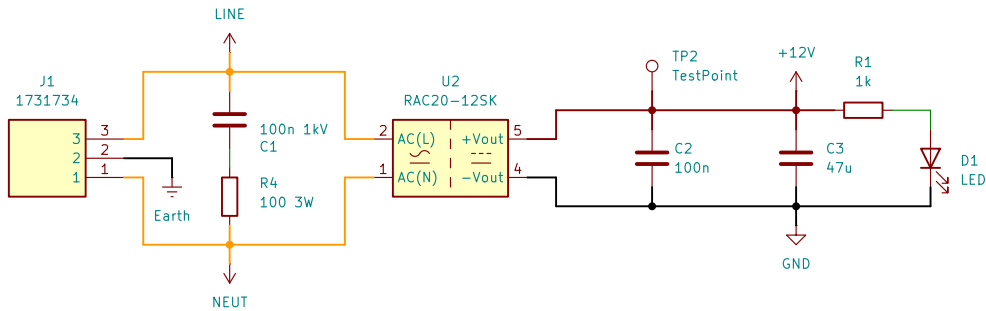


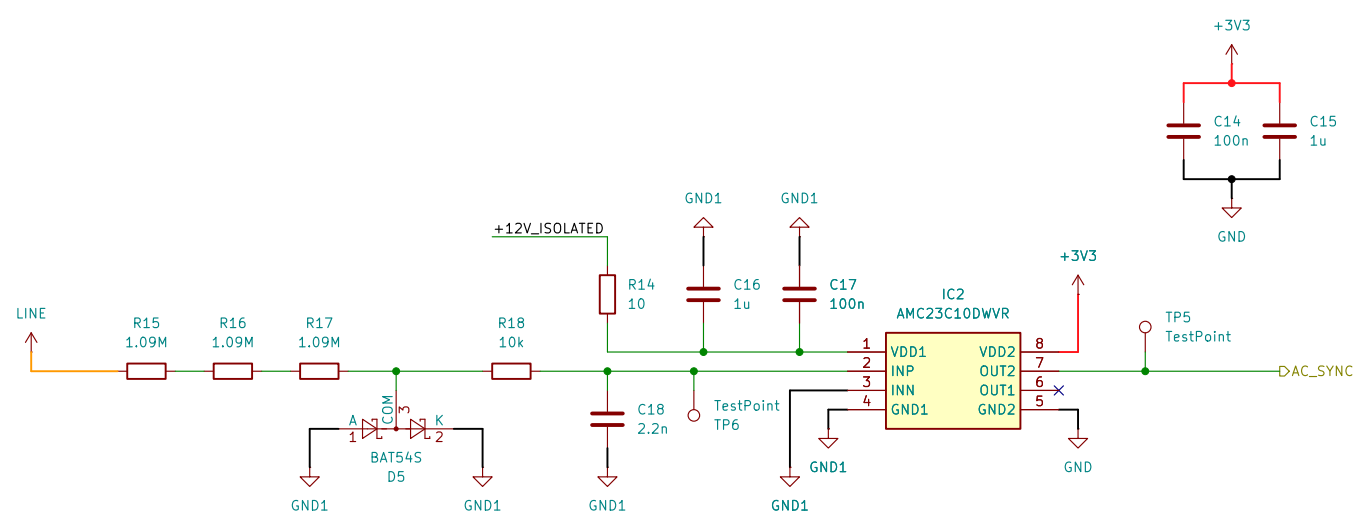


Power

230VAC power consumption:		+12V power consumption:		+5V power consumption:		+3.3V power consumption:		
-	+12V power consumption:	5.5W	-	+5V power consumption (500mA):	2.5W	-	LCD (363mA):	1.2W
-	Heaters power consumption:	4*1000W	-	+3.3V power consumption (600mA):	2.0W	-	MCU (270mA):	0.9W
-----			-	Fans: ____W		-	LED TRIAC (15mA):	0.04W
TOTAL:		4006W	-----			-	Thermocoup (600uA):	0.001W
			TOTAL: 4.5W + Fans					



AC Synchronization



Artur Twardzik

Sheet: /AC Synchronization/
File: ac_sync.kicad_sch

Title: AC Synchronization

Size: A4

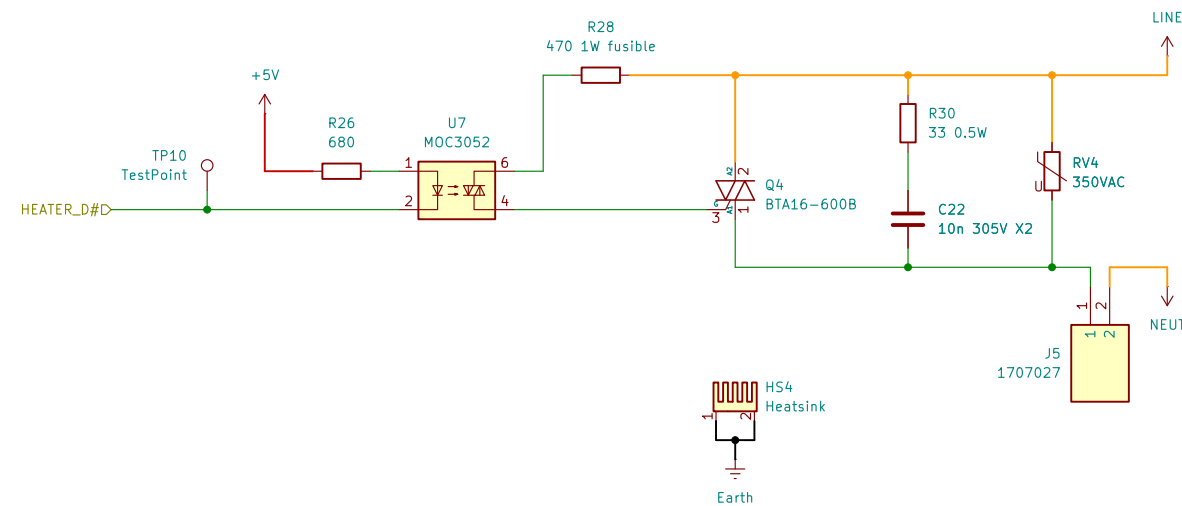
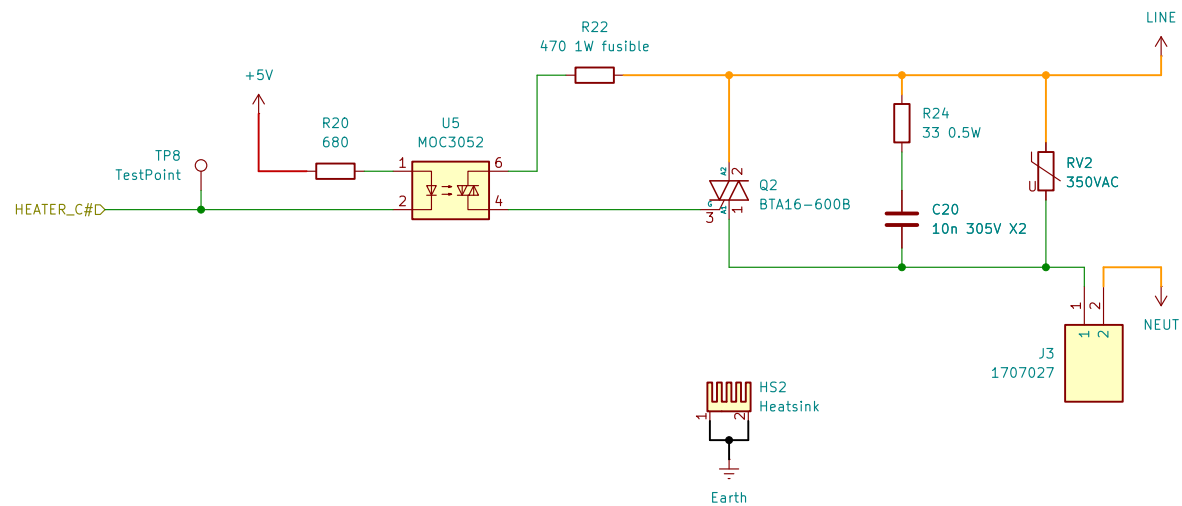
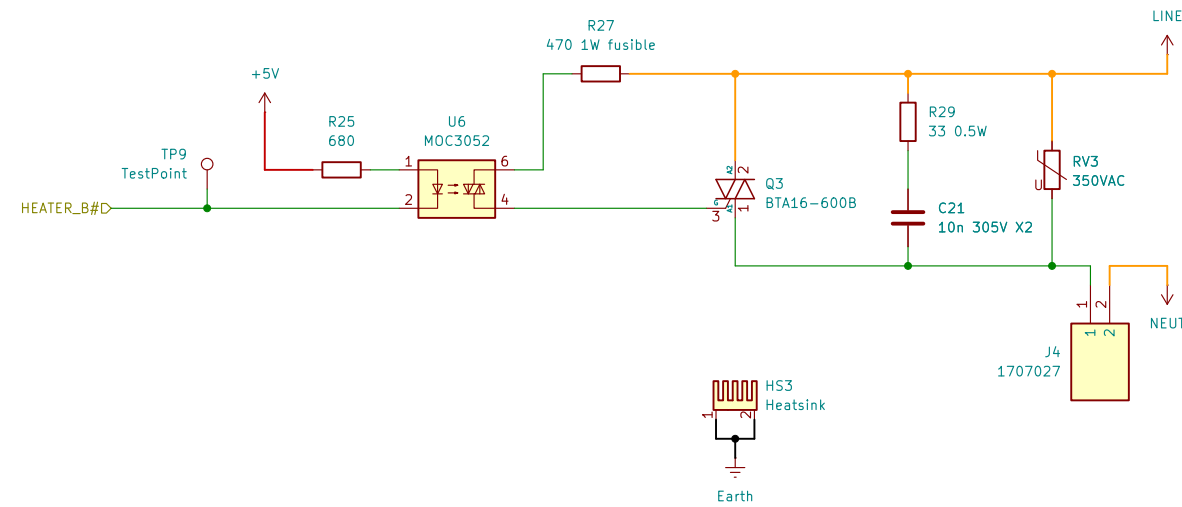
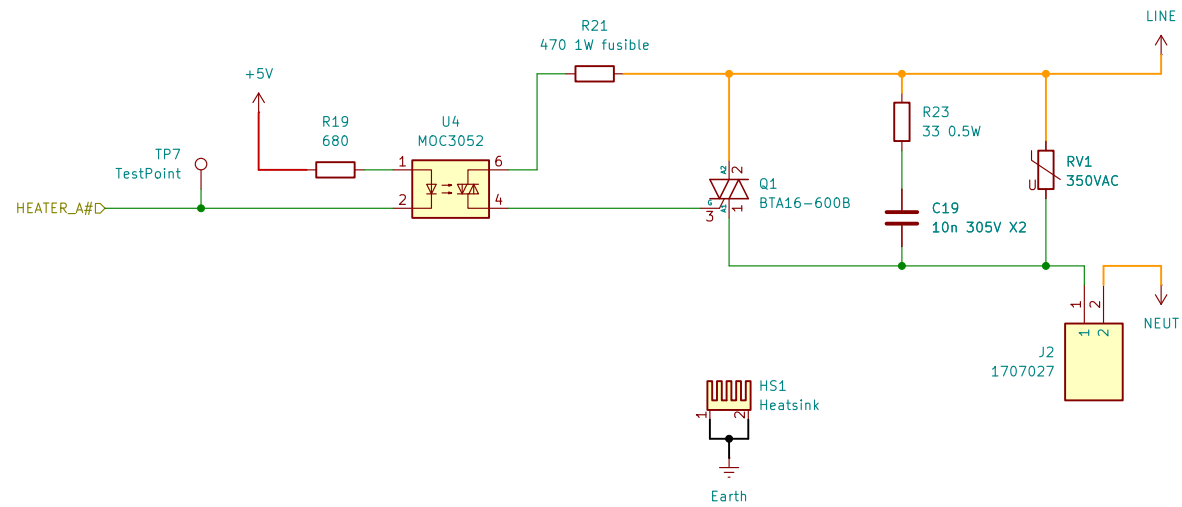
Date: 2026-09-21

Rev: 1

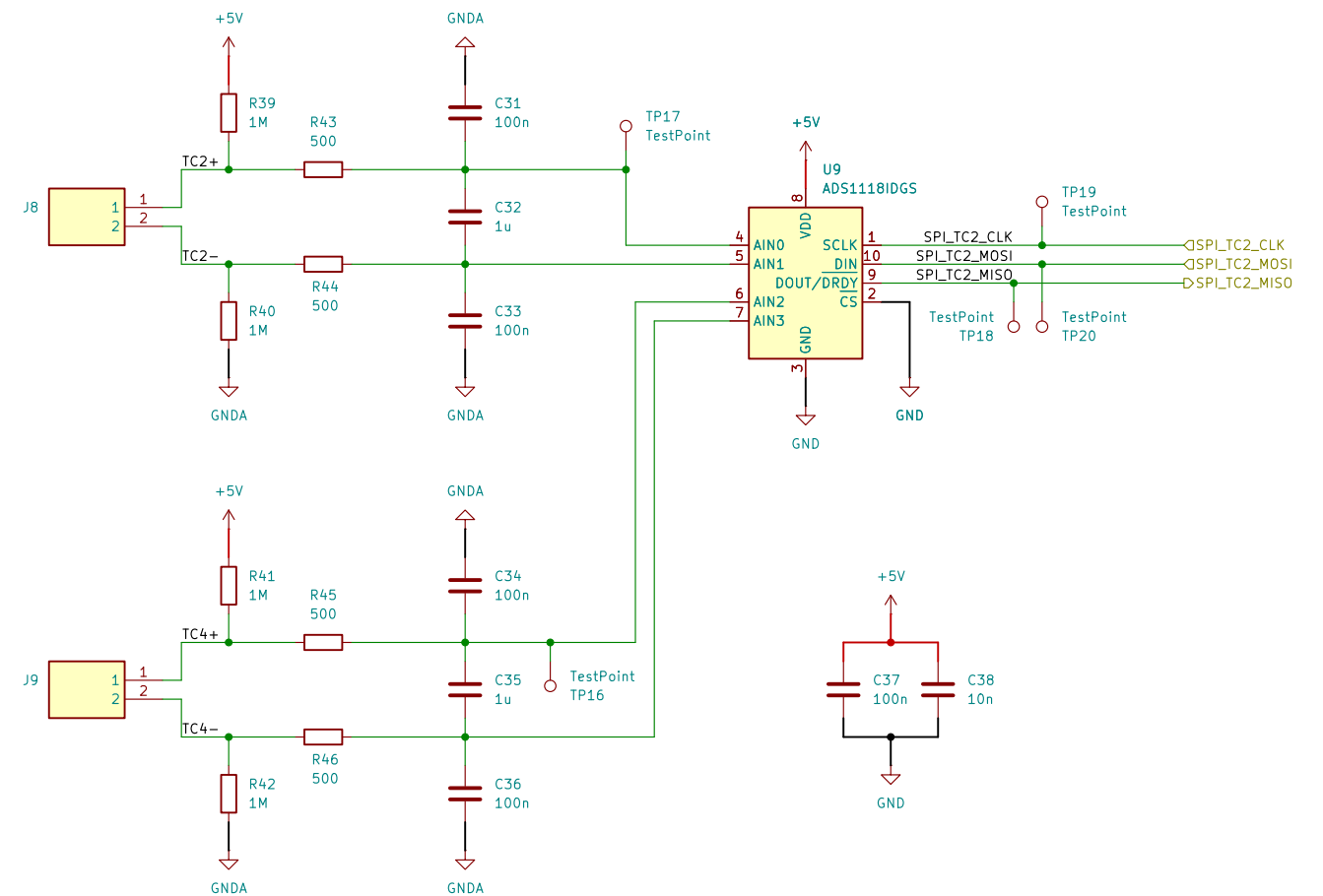
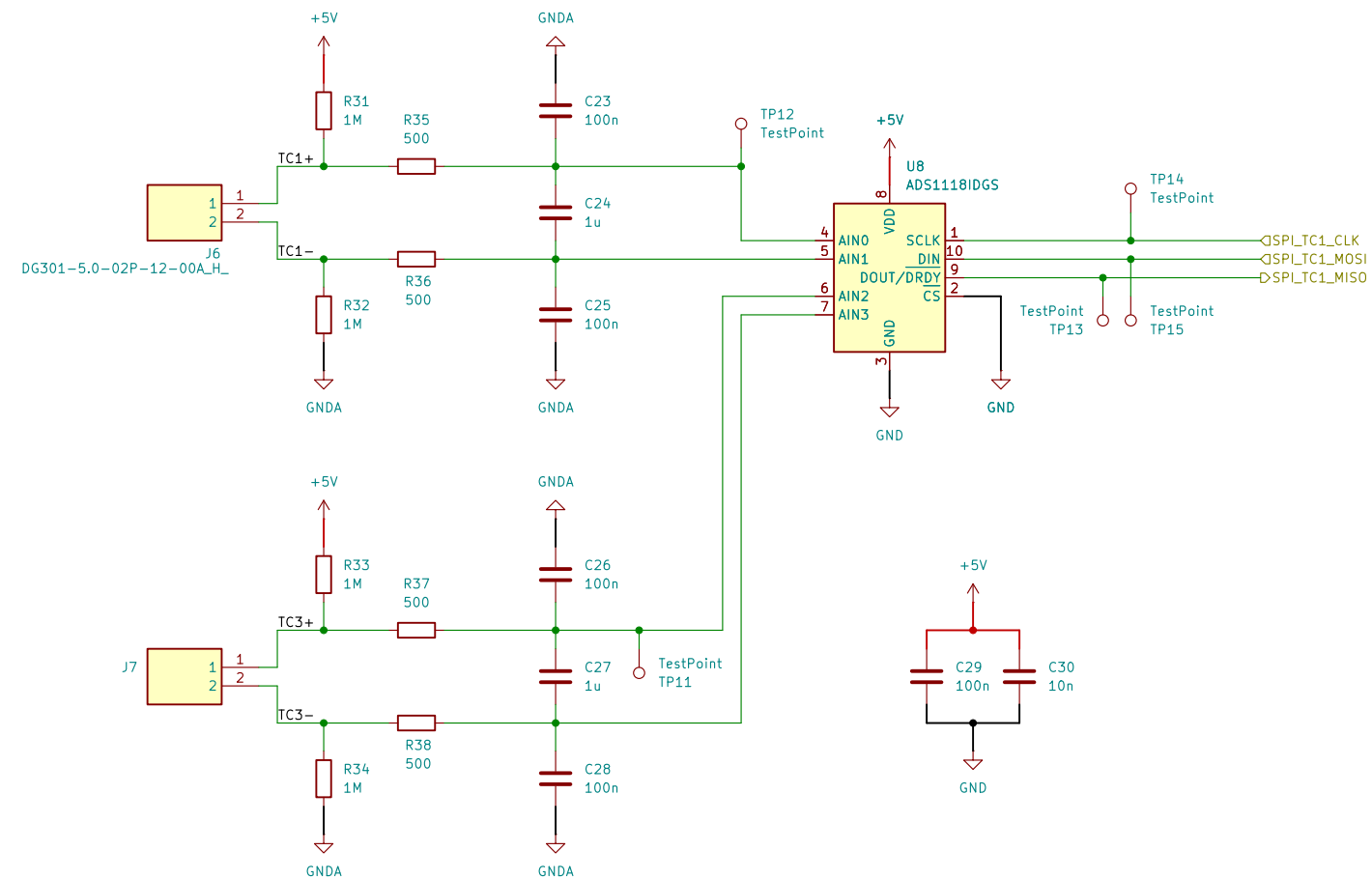
KiCad E.D.A. 10.0.6

Id: 3/16

Heaters



Thermocouples



Artur Twardzik

Sheet: /Thermocouples/
File: thermocouples.kicad_sch

Title: Thermocouples

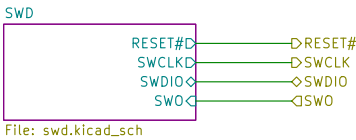
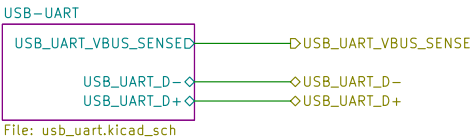
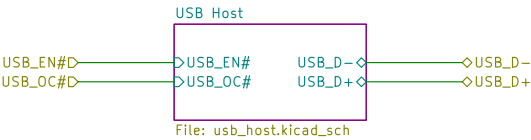
Size: A3	Date: 2026-09-21
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KiCad E.D.A. 10.0.6

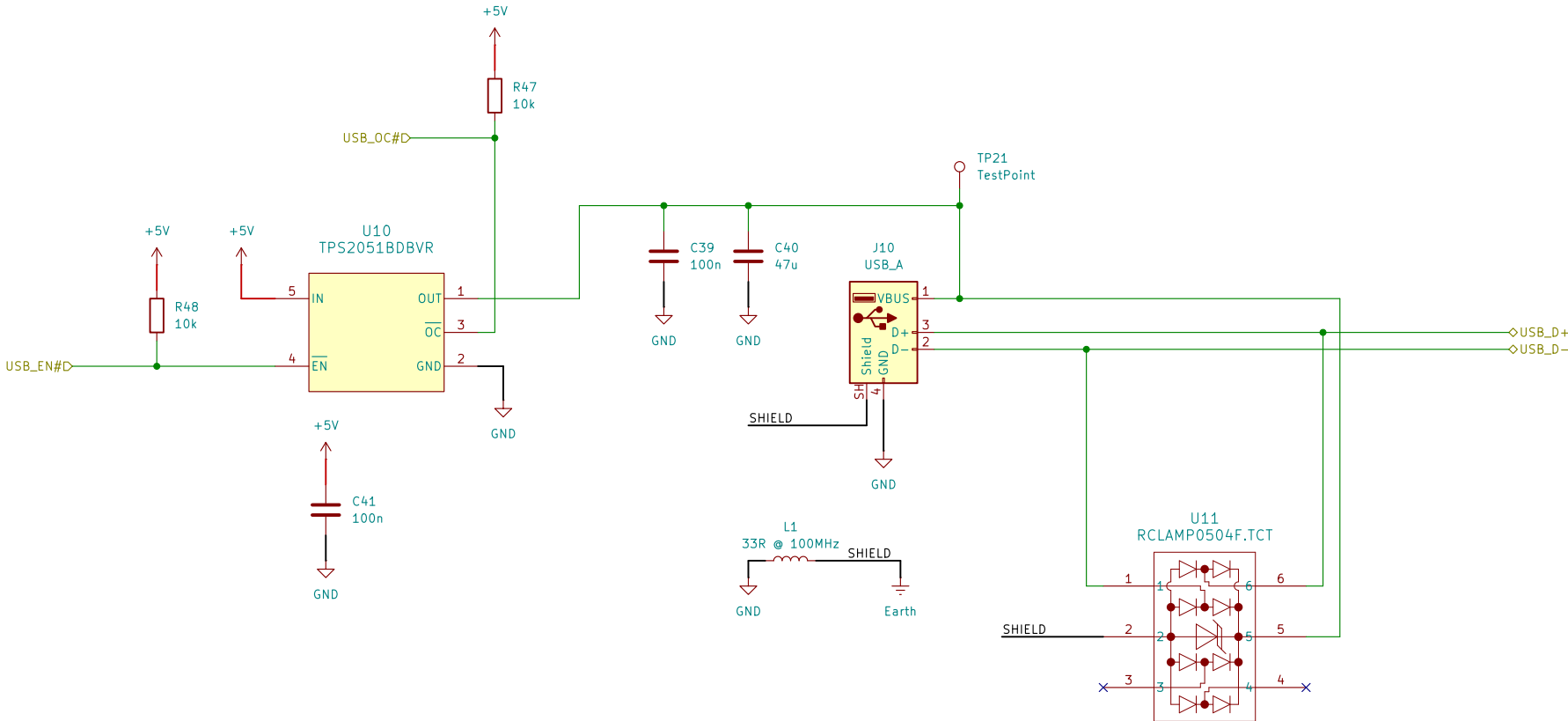
Rev: 1

Id: 5/16

Connectivity



USB



Artur Twardzik

Sheet: /Connectivity/USB Host/
File: usb_host.kicad_sch

Title: USB

Size: A4	Date: 2026-09-21
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KiCad E.D.A. 10.0.6

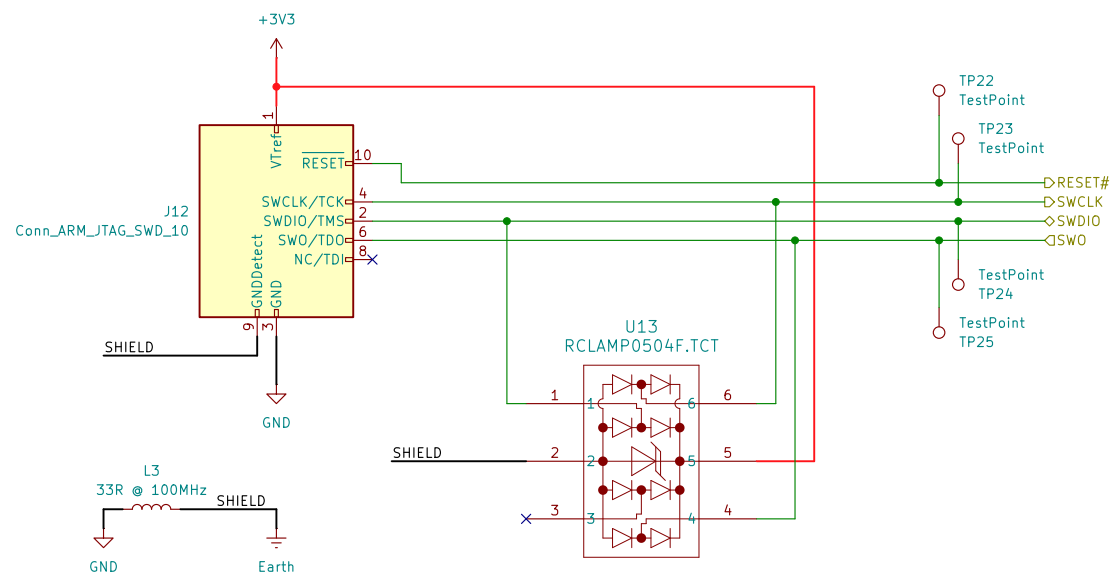
Rev: 1

Id: 7/16

The schematic diagram illustrates a USB to UART bridge circuit. It features a USB Type-C receptacle (J11) connected to a USB to UART bridge IC (U12, RCLAMP0504F.TCT). The bridge IC is connected to a UART module (U1, MAX3232CPE) via a 4-pin header (X1). The bridge IC also has a 5-pin header (X2) for additional connections. The circuit includes a 33R resistor (L2) connected to the shield of the USB Type-C receptacle. The UART module is connected to a UART module (U1, MAX3232CPE) via a 4-pin header (X1). The circuit includes a 5.1k resistor (R49) connected to the shield of the USB Type-C receptacle. The bridge IC is connected to a UART module (U1, MAX3232CPE) via a 4-pin header (X1). The circuit includes a 5.1k resistor (R50) connected to the shield of the USB Type-C receptacle. The bridge IC is connected to a UART module (U1, MAX3232CPE) via a 4-pin header (X1). The circuit includes a 33k resistor (R51) connected to the shield of the USB Type-C receptacle. The bridge IC is connected to a UART module (U1, MAX3232CPE) via a 4-pin header (X1). The circuit includes an 82k resistor (R52) connected to the shield of the USB Type-C receptacle. The bridge IC is connected to a UART module (U1, MAX3232CPE) via a 4-pin header (X1).

Rev: 1
Id: 8/16

SWD



Artur Twardzik

Sheet: /Connectivity/SWD/
File: swd.kicad_sch

Title: *SWD*

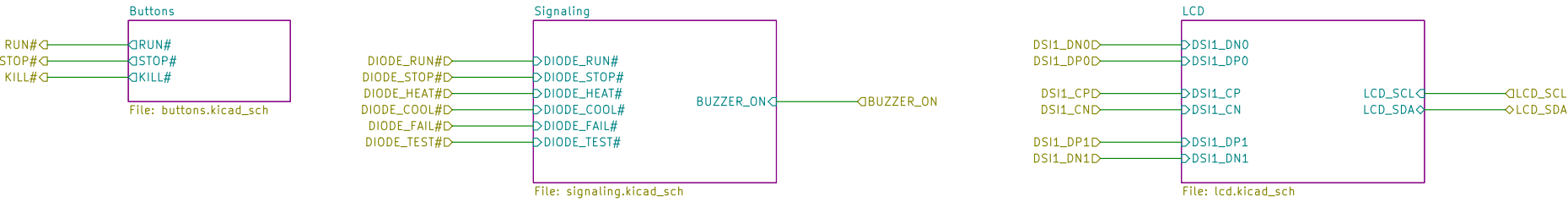
Size: A4	Date: 2026-09-21
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Size: A1	2
KiCad E.D.A. 10.0.6	

Rev: 1

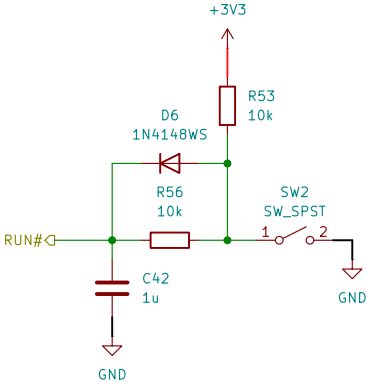
Id: 9/16

User Interface

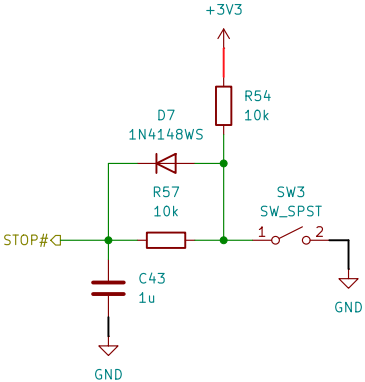


Buttons

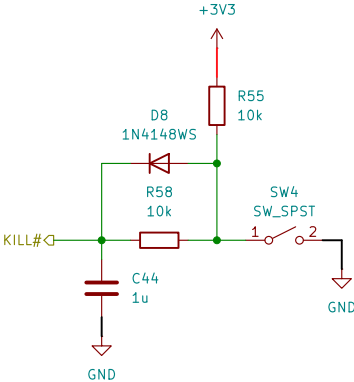
Run / Resume



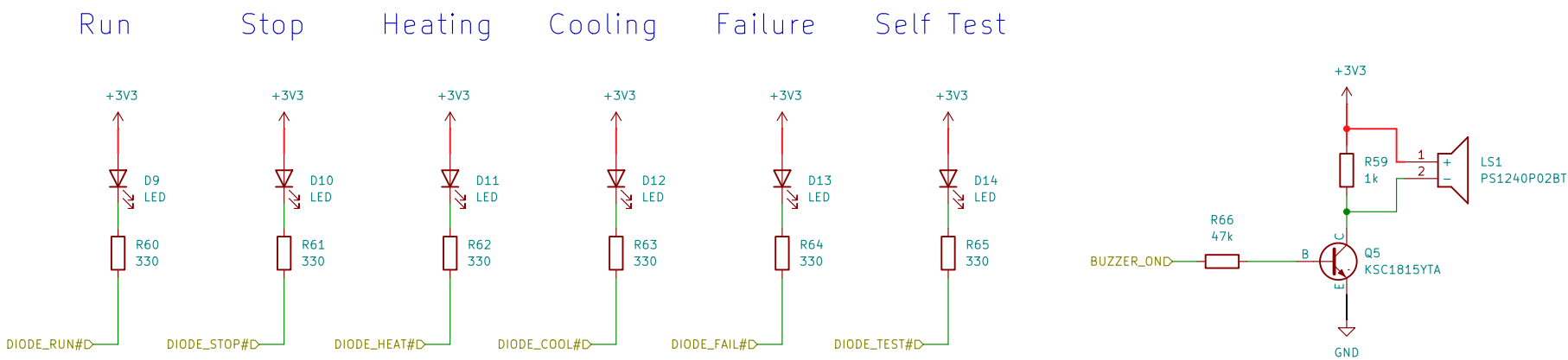
Stop & Hold



Kill



Signaling



Artur Twardzik		
Sheet: /User Interface/Signaling/		
File: signaling.kicad_sch		
Title: UI Signaling		
Size: A4	Date: 2026-09-21	Rev: 1
KiCad E.D.A. 10.0.6		Id: 12/16

The schematic diagram illustrates the electrical connections for the UI LCD module. At the center is the microcontroller unit (J13, F52Q-1A7H1-11015) with pins MP3, MP4, MP1, and MP2. The connections are as follows:

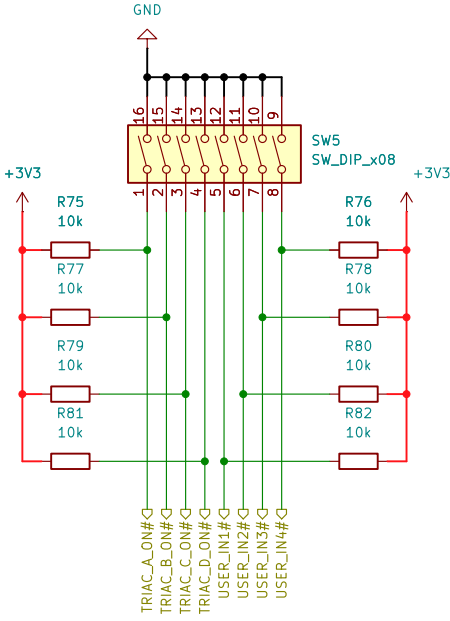
- DS18B20 Temperature Sensor:**
 - DSI1_CPD and DSI1_CND are connected to test points TP28 and TP26, respectively.
 - DSI1_DP1 and DSI1_DN1 are connected to test points TP30 and TP32, respectively.
 - DSI1_DN0 and DSI1_DP0 are connected to test points TP27 and TP29, respectively.
 - DSI1_DP1D and DSI1_DN1D are connected to test points TP33 and TP31, respectively.
- Resistors:**
 - R67, R68, R69, R70, R71, R72, R73, and R74 are 0-ohm resistors used for signal routing and termination.
- Capacitors:**
 - L4 and L5 are 33R 100MHz capacitors connected to the shield and ground.
- Other Connections:**
 - LCD_SCL and LCD_SDA are connected to test points TP31 and TP33, respectively.
 - The +3V3 supply is connected to pin 15 of the microcontroller.
 - The shield is connected to the microcontroller pins MP1 and MP2.

Metadata:

- Author: Artur Twardzik
- Sheet: /User Interface/LCD/
- File: lcd.kicad_sch
- Title: UI LCD
- Size: A4
- Date: 2026-09-21
- Rev: 1
- KiCad E.D.A. 10.0.6
- Id: 13/16

Rev: 1
Id: 13/16

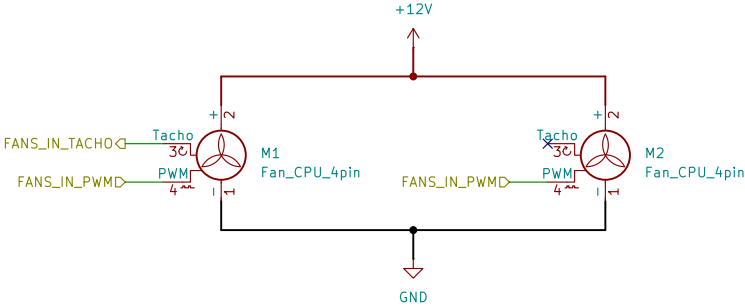
Device Configuration



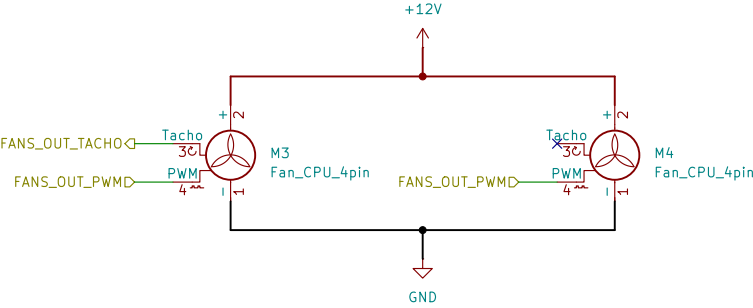
Artur Twardzik		
Sheet: /Device Configuration/		
File: config.kicad_sch		
Title: Device Configuration Switches		
Size: A4	Date: 2026-09-21	Rev: 1
KiCad E.D.A. 10.0.6		Id: 14/16

Fans

Fans In



Fans Out



Artur Twardzik

Sheet: /Fans/
File: fans.kicad_sch

Title: Fans

Size: A4 Date: 2026-09-21

KiCad E.D.A. 10.0.6

Rev: 1
Id: 15/16

MCU

